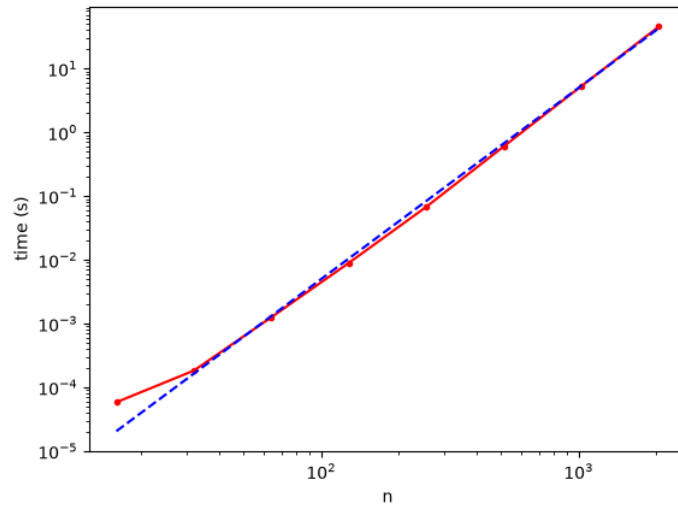


Homework 1

1. Runtime Analysis

(a) $w = 5 \times 10^{-9}$



(b)

```
1 assert a,b,c,n > 0
2 assert a,b,c <= n // set bounds that a,b,c are at most n
3
4 // get the maximum numbers of each bill for the permutation
5 max 3 buck <- min(c,n/3)
6 max 2 buck <- min(b,n/2)
7 max 1 buck <- min(a,n)
8
9 // initialize the counter
10 count <- 0
11
12
13 for (the number of max 3 buck):
14   for (the number of max 2 buck):
15     if (1. (3* the number of 3 buck for each iteration) > n
16         2. (2* the number of 2 buck for each iteration) > n
17         3. (n - (
18             (3* the number of 3 buck for each iteration)
19             +
20             (2* the number of 2 buck for each iteration)
21             ) > n
22         ) \tcc{the indentation for 3rd condition is merely for the better readability.})
```

```

23         ):
24         do nothing and move on
25     else:
26         add 1 to count
27
28 return count

```

Proof:

Let a and b be the number of 3-buck and 2-buck coins to be combined.

Let A be a 2D array that contains all sums of permutations of a AND b that those sums are always less than or equal to n .

So, the number of possible permutations is $a \times b$ and the runtime is $O(n^2)$.

If we add another condition that filters whether the value of each sum in A subtracted by n does not exceed the maximum number of 1-buck coin provided, we can calculate all the possible permutation and how many the permutations there are at the same time in n^2 times.

Hence, the asymptotic coin choice algorithm can have time complexity of $O(n^2)$.

Runtime Analysis:

1. Variable initialization: the maximum number of coins we can combine and counter $O(1)$

```

1 int max_three_buck = Math.min(c, n/3);
2 int max_two_buck = Math.min(b,n/2);
3 int max_one_buck = Math.min(a,n);
4 int count = 0;

```

2. Main Loop: $O(n^2)$

1) iterate through the maximum numbers of three-buck and two-buck coin ($O(n^2)$)

2) eliminate the possible permutations that would go out of bounds from the counter

2-1. if the given number of 3-buck coins worth more than n

2-2. if the given number of 2-buck coins worth more than n

2-3. if the given number of 3-AND-2-buck coins(permutation) worth more than n

2-4. From the permutation without 2-1,2,3., if the left-over money is greater than the maximum number of 1-buck coin.

2-5. Otherwise, put it in the counter.

3) Return the number of permutations ($O(1)$)

```

1 for (int first = max_three_buck; first >= 0; first--){
2     for (int second = max_two_buck; second >= 0; second--){
3         if(3*first > n || 2 * second > n || (3*first)+(2*second) > n || (n-(2*second+3*first)) >
         max_one_buck){
4             continue;

```

```

5         } else {
6             count++;
7         }
8     }
9 }

```

3. Return result: $O(1)$

```

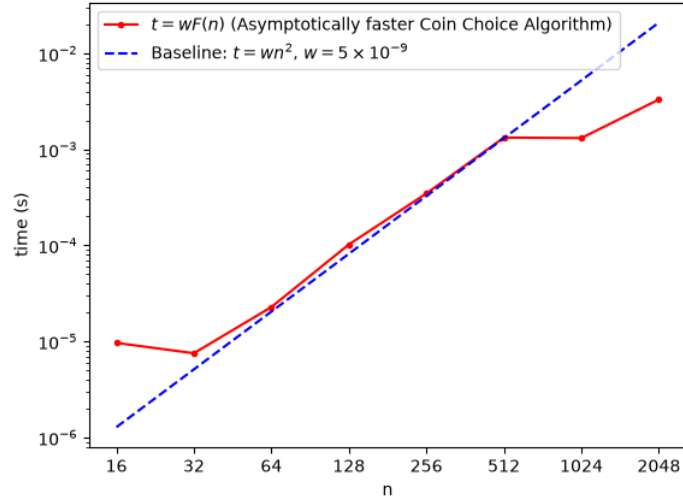
1 return count;

```

Complete Runtime = $O(1) + O(n^2) + O(1) = O(n^2)$

Therefore, the runtime of this asymptotically faster algorithm for this problem is $O(n^2)$.

(c)



Initially, w value was 5×10^{-8} as default. For part (a), I changed the exponent to -9 , which fits to the original algorithm with time complexity of $O(n^3)$. For the part (c), I did not change w value and still able to fit my asymptotically faster algorithm with time complexity of $O(n^2)$. Because w value is just an adjustment parameter to make the comparison between the baseline $t = wn^2$ and my algorithm human-friendly and my hardware setup was not changed, I did not have to change w value from part (a). One problem I encountered was the data extraction. Since I wrote the algorithm in Java and the original algorithm is in python, I had to come up with an idea that I can put into the plot in python (I don't know how to plot in Java so I had to extract data that can be read cross-linguistically.)